

## **Previous and Current Research**

My research over the past five years has followed a trajectory from single-modality PCB inspection toward multimodal, AI-assisted failure analysis of heterogeneously integrated semiconductor packages, carried out primarily at the University of Florida Security and Assurance Laboratory (SCAN Lab) under Dr. Navid Asadi with collaborations at NIST, Sandia, DHS, and industry partners.

### **Multimodal Characterization of Heterogeneously Integrated Devices**

**Location, Dates, Collaborators:** SCAN Lab, University of Florida, 2024 to present. Advisor: Dr. Navid Asadi. Frequent collaborators include Nitin Varshney, Antika Roy, Shajib Ghosh, Md Shah Imran Shovon, Charles Woychik, and Hamed Dalir.

The conceptual core of my dissertation was established in "Motivating automated multimodal failure analysis for heterogeneously integrated devices" (Craig et al., SPIE 2024), which argues that heterogeneous material stacks in advanced packages exceed the sensitivity of any single non-destructive modality and motivates a fused-inspection framework combining X-ray, acoustic, terahertz, and optical measurements with AI-based fusion. I have since built out this workflow using the SCAN Lab's Sigray Apex Hybrid X-ray CT and laminography system, Nordson Sonoscan SAM, Protemics TeraCube near-field THz-TDS, and optical microscopy, investigating relationships between imaging physics, material properties, and defect observability. I am co-inventor on two 2026 US patent applications formalizing outputs: "Explainable AI metrics for IC packaging inspection" and "Image reconstruction with multimodal fusion and physics-informed neural network."

### **NIST Collaboration on Multimodal X-Ray Characterization of Through-Glass Vias**

**Location, Dates, Collaborators:** National Institute of Standards and Technology, Spring 2026. SCAN Lab collaboration with NIST staff Dr. Jessie Zhang.

I collaborated with NIST on multimodal X-ray characterization of advanced through-glass via (TGV) and redistribution layer (RDL) structures for heterogeneous integration. I designed and executed imaging workflows

using X-ray CT and laminography across multiple wafer-scale and die-scale samples with voxel resolutions from 5 micrometers to 300 nanometers, characterized high-aspect-ratio TGV structures with via pitches spanning 100 to 400 micrometers, and investigated void formation, copper pullout, delamination, interface defects, and process-induced variability across bonded and freestanding wafer configurations. I developed region-of-interest and montage acquisition strategies to optimize throughput, defect detectability, and cross-sectional correlation, and contributed comparative analyses between X-ray modalities that quantified defect observability limitations and informed non-destructive inspection methodology.

### **Multimodal Characterization Dataset for HBM3 Memory Systems**

**Location, Dates, Collaborators:** SCAN Lab, University of Florida, 2025 to present. Target venue: ISTFA 2026. Collaborators within the SCAN Lab group.

I am developing a large-scale multimodal characterization dataset for NVIDIA H100 HBM3 memory systems, integrating X-ray CT, X-ray laminography, and scanning acoustic microscopy for advanced packaging inspection and defect analysis. The work performs non-destructive imaging of HBM interconnects, microbumps, TSVs, substrate interfaces, and multilayer package structures, and generates and curates thousands of annotated defect and interconnect features. The dataset is being designed to support agentic AI workflows for semiconductor inspection, automated failure analysis, adaptive defect localization, and advanced packaging metrology research, and it serves as a benchmark for the multimodal fusion methods central to my dissertation.

### **Sandia National Laboratories LDRD Semiconductor Assurance Project**

**Location, Dates, Collaborators:** Sandia National Laboratories LDRD collaboration with SCAN Lab, 2025 to present. Funding: 130,000 US dollars.

I contribute to a Sandia LDRD-funded initiative on semiconductor assurance and material-aware X-ray characterization. My contributions include multimodal workflows integrating multi-energy X-ray CT and laminography with spectral reconstruction to analyze material properties and defect behavior, methods for

reconstructing effective atomic number and electron density within 3D X-ray volumes, and software pipelines for material-aware 3D PCB reconstruction including GERBER-to-material representations, spectral CT integration, and layer-alignment visualization.

### **PCB Assurance and Netlist Extraction**

**Location, Dates, Collaborators:** SCAN Lab, University of Florida, 2023 to 2025. Frequent co-authors include Nitin Varshney, Antika Roy, Shajib Ghosh, Jonathan Pearson, Sanjeev J. Koppal, and Charles Woychik.

My earliest doctoral work centered on non-destructive inspection of printed circuit boards and produced the methodological foundation on which the current multimodal work rests. In "Advancing PCB assurance towards netlist extraction with the integration of X-ray imaging and semi-supervised learning techniques" (RAPID 2024) I demonstrated that semi-supervised deep learning can substantially reduce the labeled-data burden for PCB component detection under X-ray. I extended this to a multimodal setting in "Multi-modal printed circuit board netlist extraction with X-ray and optical imaging" (SPIE 2024), the direct predecessor of my packaging-scale multimodal work. In parallel I developed an optical PCB-interconnect inspection pipeline presented at ISTFA 2024 and extended into the journal article "AI-Driven Framework for Generalized Optical Inspection of Printed Circuit Board Interconnects" (Journal of Failure Analysis and Prevention, 2025), and reported the high-throughput X-ray extension in "Deep Learning-Driven X-Ray Analysis for High-Throughput Manufacturing" (RAPID 2025).

### **Terahertz Time-Domain Spectroscopy for Integrated-Circuit Identification**

**Location, Dates, Collaborators:** SCAN Lab, University of Florida, 2023 to 2024. Collaborators: Chengjie Xi, William Mitchell, Nitin Varshney, Shajib Ghosh.

As part of broadening the modality set beyond X-ray and optical imaging, I first-authored "Terahertz time-domain spectroscopy (THz-TDS) fingerprinting for integrated circuit (IC) identification in tracking and tracing applications" (SPIE Terahertz Emitters, Receivers, and Applications XV, 2024), which established near-field

THz-TDS as a distinguishing modality for IC identification and provenance. This work introduced the Protemics TeraCube capability into my multimodal workflow and provides the terahertz contribution to the fusion framework.

### **Program Leadership and Broader Contributions**

**Location, Dates, Collaborators:** SCAN Lab, University of Florida, 2023 to present.

I serve as technical lead for a multi-million-dollar NSF Engines-funded semiconductor R&D initiative, leading development of multimodal datasets and AI-assisted failure-analysis workflows. I have also co-authored numerous SCAN Lab papers (SPIE, ECTC, RAPID, PAINE, and Microscopy and Microanalysis venues, 2023 to 2026) spanning explainable-AI metrics, PINNs for X-ray forward models, SAM adaptation for PCB X-ray, counterfeit-IC classification, and VLM/LLM assistants for inspection.

### **DHS HS-POWER Program on Automated Microelectronics Assurance**

**Location, Dates, Collaborators:** US Department of Homeland Security, Science and Technology Directorate, 2023 to 2025 (two program years). Mentor: Dr. Pauline Paki, Senior Scientific Advisor. SCAN Lab co-authors: Liton Kumar Biswas, Nitin Varshney, Antika Roy, Clark Duke Godinez.

Over two program years I contributed to the DHS Homeland Security Professional Opportunities for Student Workforce to Experience Research (HS-POWER) program under Dr. Pauline Paki on non-destructive automated microelectronics assurance across X-ray CT/laminography, SAM, and optical inspection, and co-authored "Advanced packaging raw materials assurance by characterization" (Biswas et al., SPIE 2025). Across two summers I also mentored three bachelor's- and master's-level student interns to publishable outputs on assurance and counterfeit detection.